English 日本語

Sho Sakai

PH.D. STUDENT IN MATHEMATICS
UNIVERSITY OF TSUKUBA



↗ BLOG ↗ CV

ssakai[at]math.tsukuba.ac.jp
github.com/ShoShohh
@simplesho_CLT
linkedin.com/in/ssakai0403
Google Scholar
orcid.org/0009-0005-3408-4622
↗ Kaggle Profile

My name is **Sho Sakai**, a Ph.D. student in the Mathematics Degree Program (Graduate School of Science, Degree Programs in Pure and Applied Sciences), University of Tsukuba. I am a **JST BOOST Next-Generation AI Human Resource Development Program** fellow, supported to pursue doctoral research in statistics and related fields (see **Education**, **Fellowships**, and **Work Experience** below for details).

I am interested in **high-dimensional statistical analysis**—topics such as dimensionality reduction and inference for principal component regression in high dimensions—as well as applications to real data and decision support. Personally, I am also curious about finance: how statistical structure and inference can connect to the **quality of decisions** in markets, including market data, risk measurement, and valuation or allocation—not as a trading game, but as careful evaluation under uncertainty.

Alongside my degree I work as a **contract analyst** at **Nospare Inc.**, contributing to statistical and DX-oriented consulting, hands-on data analysis and machine learning, and education programs.

I host the **podcast "Data Science LG: Learning Together in Statistics and Data Science"**, where we explore statistics, machine learning, and academic careers from the perspective of students and researchers. The podcast is available on **Apple Podcast**, **Spotify**, **YouTube**, and **Amazon Music**. Through the **Data Science Learning Group** community, I plan and run study sessions on statistics, machine learning, deep

learning, and related topics. For more on the study groups and podcast, please see the **Community / Podcast** section below.

As a **Bone Marrow Bank youth ambassador**, I help raise awareness of **hematopoietic stem cell transplantation**: I share my experience as a donor, promote donor registration, and take part in talks and media appearances. Further links and write-ups are in **Outreach / Public Engagement** below. Through these activities I hope to connect healthcare and society.

I share **my research findings and code** on **GitHub**, and post about blood donation, hematopoietic stem cell transplantation, data science, education, and research on **note**.

INTERESTS

High-Dimensional Statistics | HDLSS (High-Dimensional Low Sample Size) | Machine Learning | Dimensionality Reduction (PCA, PLS, CCA, MCA, ICA) | Representation Learning | generative Model | Statistical Causal Discovery (LiNGAM) | Text Embedding | Finance | Open Science

NEWS

Nov. 1, 2026

- Internship — Added Mitsubishi UFJ Trust Investment Technology Institute (MTEC), Oct. 2026.
- Qualification / Test Score — Added the Matsuo–Iwasawa Lab Deep Generative Models (Deep Learning Applied Course) 2026 Summer certificate.
- Community / Podcast — Added guest Fuyuto Miyake (episode #48).
- Contact — Added ORCID.

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Portfolio

SKILL

See full profile on LinkedIn

SPEECH

Sho Sakai, Kazuyoshi Yata, and Makoto Aoshima, "Exploring Principal Component Regression in High-Dimensional Data", *SPRING and BOOST Selected Students Research Presentation*, Ibaraki, Japan, Poster Presentation, March 12-13, 2026.

View Program

Sho Sakai, Kazuyoshi Yata, and Makoto Aoshima, “Exploring Principal Component Regression in High-Dimensional Data: Hypothesis Testing for PCR Coefficients & Steps Toward Prediction-Error Minimisation”, *Seminars by Alumnae/Alumni of Kagoshima University on their Recent Achievements 2026*, Kagoshima, Japan, Oral Presentation, March 9-10, 2026.
View Program

Sho Sakai, Kazuyoshi Yata, and Makoto Aoshima, “Exploring Principal Component Regression in High-Dimensional Data”, *The 20th Japan Statistical Society Spring Meeting*, Kyoto, Japan, Poster Presentation, March 7, 2026.
View Program

Sho Sakai, Kazuyoshi Yata, and Makoto Aoshima, “Exploring Principal Component Regression in High-Dimensional Data: Hypothesis Testing for PCR Coefficients & Steps Toward Prediction-Error Minimisation”, *ISACT 2026*, Tokyo, Japan, Poster Presentation, February 16-17, 2026.
View Program

Sho Sakai, Kazuyoshi Yata, and Makoto Aoshima, “Exploring Principal Component Regression in High-Dimensional Data”, *異分野異業種研究交流会2025*, Tokyo, Japan, Poster Presentation, October 25, 2025.
View Program

Sho Sakai, Kazuyoshi Yata, and Makoto Aoshima, “Exploring Principal Component Regression in High-Dimensional Data: Hypothesis Testing for PCR Coefficients & Steps Toward Prediction-Error Minimisation”, *Statistics Summer Seminar 2025*, Kagawa, Japan, Poster Presentation, August 4-6, 2025.
View Program

Sho Sakai, Kazuyoshi Yata, and Makoto Aoshima, “Hypothesis testing for PCR coefficients in high-dimensional data”, *The Mathematical Society of Japan Annual Meeting*, Tokyo, Japan, Oral Presentation, March 21, 2025.
View Program

Sho Sakai, Kazuyoshi Yata, and Makoto Aoshima, “Hypothesis testing for PCR coefficients in high-dimensional data”, *Seminars by Alumnae/Alumni of Kagoshima University on their Recent Achievements 2025*, Kagoshima, Japan, Oral Presentation, March 11, 2025.
View Program

EDUCATION

2025 – 2028 (expected)



University of Tsukuba
Ph.D. in Mathematics, Graduate School of Science, Degree Programs in Pure and Applied Sciences

Advisor: Aoshima Laboratory

2023 – 2025



University of Tsukuba
M.Sc. in Mathematics, Graduate School of Science, Degree Programs in Pure and Applied Sciences

Advisor: Aoshima Laboratory

2019 – 2023



Kagoshima University
B.Sc. in Mathematics and Informatics, Faculty of Science, Mathematics and Informatics Program

Advisor: Yoshida Laboratory (2022–2023)

2016 – 2019

Ehime Prefectural Matsuyama Kita High School
Graduated (General Course)

WORK EXPERIENCE

Jan. 2026 – present

Analyst (Contract; transitioned from internship) — Nospare Inc.

Independent contractor after completing the internship (Dec. 2024 – Dec. 31, 2025), which was limited to the education business. The contract role spans consulting (management, DX, and statistical advisory), solutions-oriented data analysis and machine learning, and continued involvement in education—on-demand programs such as EDRIGHT and statistics-focused seminars for practitioners—plus analysis, modeling, and implementation support for client needs.

Mar. 2019 – Oct. 2024

Part-time work — Food service (front of house & kitchen), liquor retail, logistics & warehousing

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INTERNSHIP

Oct. 2026

[Mitsubishi UFJ Trust Investment Technology Institute Co., Ltd. (MTEC)]

Internship at an MUFG Group research institute that addresses practical problems in financial services by combining mathematical and information sciences, and provides financial solutions for real operations.

Nov. 2025

[Mizuho–DL Financial Technology Co., Ltd.]

Internship at a company engaged in management and risk management enhancement for financial institutions, research and development and utilization of data analytics technologies, and development of investment and asset management methods.

Aug. 2025 (3 days)

[Mizuho Research & Technologies, Ltd.]

AI field internship as an R&D specialist. Participation in an image recognition AI workshop focused on manufacturing inspection. Experience with practical AI challenges such as limited anomaly data and unstable imaging conditions, and hands-on verification and analysis with image recognition AI.

Dec. 2024 – Dec. 31, 2025

[Nospare Inc.]

Internship focused on the education business only—supporting on-demand programs such as EDRIGHT, seminar-related operations, and other educational initiatives. The internship concluded on Dec. 31, 2025, followed by transition to an independent contract role with a broader scope.

FELLOWSHIPS

Apr. 2025 – Mar. 2028

Next-Generation AI Human Resource Development Program Fellow — JST BOOST initiative, University of Tsukuba

Selected for the University of Tsukuba's Next-Generation AI Talent Development Program, "Project for Interdisciplinary Next-Generation AI Innovative Human Resource Development". This initiative, part of a national strategy to foster doctoral students, supports 600 students across Japan. The University of Tsukuba has a quota of 12 students, and I was chosen as one of them through a highly competitive internal selection process. Receiving support for living expenses and research funds, I am able to fully dedicate myself to theoretical and applied research in high-dimensional statistics. I plan to actively pursue collaborations with fields such as bioinformatics, materials science, and astronomy.

PDF

RESEARCH ASSISTANT

Oct. 2023 - Nov. 2024

University laboratory

TEACHING & ACADEMIC SERVICE

Oct. 2023 – Mar. 2025

Teaching Assistant — University of Tsukuba

Courses: Statistical Exercise (Fall 2023, Fall 2024), Computer Exercise (Fall 2023, Fall 2024), Computer Mathematics I (Spring 2024), Linear Algebra I (Spring 2024).

Apr. 2024 – present (Chair: Apr. 2025 – Mar. 2026; Vice Chair: Apr. 2026 – present)

Vice Chair & Technical Assistant — User Committee, Department of Mathematics, University of Tsukuba

Served as technical assistant for the graduate student user committee under the Department of Mathematics, University of Tsukuba, providing technical support for the management and maintenance of the department's computing environment. Held the position of chair from Apr. 2025 to Mar. 2026 and vice chair from Apr. 2026 to present.

Dec. 2023 / Dec. 2024

International Symposium Staff — International Symposiums on Large Complex Data

Contributed to the 2023 International Symposium on Recent Advances in Theories and Methodologies for Large Complex Data and the 2024 International Symposium on Theories, Methodologies and Applications for Large Complex Data.

Jul. 2023 / Jul. 2024

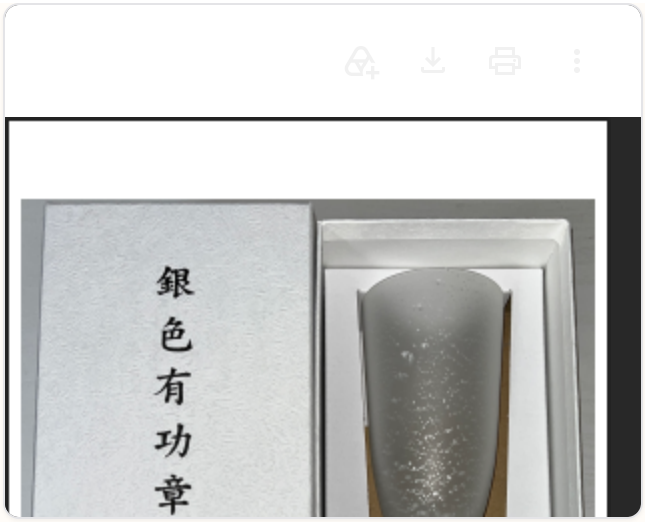
Staff, Mathematics Trial Program — University of Tsukuba

Served as staff for the undergraduate mathematics experience program in July 2023 and July 2024.

AWARD

Dec. 1, 2025

Silver Medal of the Japanese Red Cross（Japanese Red Cross）



Open PDF in new tab

Bone Marrow Donation Certificate of Appreciation（Minister of Health, Labour and Welfare)



Open PDF in new tab

Oct. 28, 2020

Common Education Center Director's Award（Kagoshima University）



Open PDF in new tab

COMMUNITY / PODCAST

Community "Data Science Learning Group" • Podcast "Data Science LG: Learning Together in Statistics and Data Science"

2024 - present

In the Data Science Learning Group community, I plan and run study sessions on statistics, machine learning, deep learning, and related topics, with participants from diverse backgrounds including students and industry practitioners. I am also advancing the design and public release of operational workflows for study groups. You can check currently open and ongoing sessions below. If the application form does not work, please contact us via social media DM to express your interest in joining.

Recruitment Progress

I host the podcast "Data Science LG: Learning Together in Statistics and Data Science", where we explore topics such as statistics, machine learning, and academic careers from the perspective of students and researchers. The podcast is available on Spotify, Apple Podcast, YouTube, and Amazon Music.

For an overview of our podcast activities and related information, please visit the Notion page below:

Apple Podcast
Spotify
YouTube
Amazon Music
Podcast Activity Overview
News
Audience Data
Note
Guest Confirmation Checklist
Conference Review Guidelines



Cast / Guest List

Titles and affiliations are as of the time of recording.
Apple Podcast links are provided for each entry; the show is also available on Spotify, YouTube, and Amazon Music.
* indicates guests who have appeared in multiple episodes. The episode link is to their first appearance; you can find other episodes by searching for their name on each platform.

津志田 侑弥さん*

立正大学データサイエンス学部 2021年度入学

LinkedIn
🔗

中山 優吾 さん*

企業研究員

HP
LinkedIn
🔗

岩永 悠希さん

神戸大学大学院経済学研究科 修士２年

HP
🔗

上野 孝斗 さん*

滋賀大学大学院データサイエンス研究科 修士2年

X
🔗

久保 知生さん

dip株式会社

X
Link
🔗

司馬 博文さん*

総合研究大学院大学 先端学術院 統計科学コース 五年一貫博士課程 2年
→総合研究大学院大学統計科学コース5年一貫D3（2025年度より）

発表資料
X
HP
🔗

平木 大智さん*

東京大学経済学研究科統計コースD1（2025年度より）

X
LinkedIn
発表内容の論文
🔗

薗部 成輝 さん

東京理科大学 創域理工学研究科 情報計算科学専攻修士２年

X
公開されている論文
論文について
🔗

加藤 真大さん*

みずほ第一フィナンシャルテクノロジー データアナリティクス技術開発部
東京大学 博士後期課程

X
LinkedIn
Webサイト
🔗

長谷川 弘貴さん*

筑波大学大学院理工情報生命学術院システム情報工学研究群サービス工学学位プログラム 修士2年

LinkedIn
HP
🔗

井下 敬翔さん

関西大学 大学院 商学研究科 商学専攻 博士後期課程

HP
Researchmap
X
LinkedIn
🔗

戸簾 隼人さん*

滋賀大学 大学院 データサイエンス研究科 博士後期課程

HP
LinkedIn
Researchmap
🔗

MC: 酒井 彰*

筑波大学 大学院 博士後期課程
全ての回に出演しています。編集やゲストへの連絡などの運営を担当しています。

Nospare Student Community 運営
X
LinkedIn
HP

MC: 北野 優斗*

早稲田大学 大学院 修士課程

Nospare Student Community 運営
X
🔗

MC: 段 暁然*

EPSRC CDT in Statistics and Machine Learning at Imperial College London and the University of Oxford (博士課程)

X
🔗

佐野 和幸さん

LINEヤフー株式会社 データサイエンティスト

X
LinkedIn
🔗

金髙 右京さん

Sansan株式会社 研究開発部 研究員

X
Eightデジタル名刺
🔗

井口 亮さん

みずほ第一フィナンシャルテクノロジー
データアナリティクス技術開発部長

LinkedIn
🔗

陳 鈺涵さん

みずほ第一フィナンシャルテクノロジー
ファイナンシャルエンジニア

HP

🔗

増田 伊吹 さん*

筑波大学大学院 情報理工学位プログラム D1

LinkedIn

🔗

須山 敦志さん

Aspire Analytics合同会社 CEO

HP
X
note

🔗

種村 賢飛さん

東京大学工学部計数工学科B4
株式会社松尾研究所 AI ソリューション開発事業 AI エンジニア

LinkedIn
学生コミュニティ UTokyo Finance Community
UTFC Xアカウント

🔗

三宅 冬人さん

竹田綜合病院 内科／AIエンジニア

X
note
LinkedIn

🔗

Conference Reviews

Apple Podcast links are provided for each entry; the show is also available on Spotify, YouTube, and Amazon Music.

計算技術による学際的統計解析ワークショップ（#14, 15）

2025年6月11日

🔗 エピソード

第19回日本統計学会春期集会（#17, 18）

2025年6月11日

🔗 エピソード

日本数学会2025年度年会（#19）

2025年6月11日

🔗 エピソード

日本応用数理学会2025年度年会（#41）

2025年9月21日

🔗 エピソード

白金鉱業Meetup vol.20 効果検証編（#42）

2025年10月11日

🔗 エピソード　🔗 note

NeurIPS 2025、CMStat 2025、みずほ、統計サマーセミナー 2025、統計関連学会連合大会 2025、異分野異業種研究交流会 2025（#44）

2025年12月28日

🔗 エピソード

Podcast Reviews

Apple Podcast links are provided for each entry; the show is also available on Spotify, YouTube, and Amazon Music.

#26 半年間の振り返りと今後の構想、データサイエンス系Podcastのおすすめ

2025年6月11日

🔗 エピソード

#44 NeurIPS & CMStat 2025振り返りレポ、みずほR&T・みずほ第一FTのインターン（おまけ：統計サマセミ、統計連合大会、異分野交流会、2025総括も） w/ 筑波大 増田さん

2025年12月28日

🔗 エピソード

Articles

(Jul. 23, 2026) 🎧 **Batch-generating podcast deliverables with an md template + audio — reproducible AI workflows**

🎧 Podcast公開物は「mdテンプレ＋音声」で一括生成する ～再現...



追記 Riversideでは、Magic Clipsという機能で動画の切り抜きや、他には編集などを自動で行えるようになっています！mdには不要な内容がところどこ...

♡ 12



Sho Sakai
2026/07/23 23:00

(Jan. 31, 2026) 🎧 Podcast production when you're short on time but care about quality – Full use of AI

🎧 時間はないけどこだわりたいPodcast作成 ～AIフル活用術～



追記（2026-07-23） 公開物まわり（タイトル・概要・チャプター・Shorts・SNS告知など）は、いまは mdテンプレ＋音声で一括生成 する運用に切り...

♡ 13



Sho Sakai
2026/01/31 01:48

(Oct. 10, 2025) 🎧 Shirokane Kogyo Meetup vol.20 Effect Verification Report w/ LINE Yahoo Corporation Sano Kazuyuki, Sansan Corporation Kindaka Ukyo

🎧 白金鉱業Meetup vol.20 効果検証編レポ w/ LINEヤフー株式...

2025年9月18日に 「白金鉱業 Meetup vol.20 効果検証編」 が開催されました。このMeetupは、株式会社ブレインパッドの社員の方々が中心となり、...



♡ 16



Sho Sakai
2025/10/10 23:09

Tsukuba Graduate Students' Network • Podcast 'Graduate Student Café – University of Tsukuba Branch'

2024 - 2025

I was a general member of this graduate student community and started a podcast initiative within the group.

[Community Website] [Spotify]

OUTREACH / PUBLIC ENGAGEMENT

Blood donation

2018 – present

Achieved 70 blood donations and received the silver merit badge.

(August 13, 2025) **Blood Donation RTA: Silver Merit Badge Achievement (70 donations reached) Report**

献血回数RTA 銀色有功章獲得！(70回到達)レポート ＋ 献血のススメ



🩸 記録タイム：９年４ヶ月10日 16歳の誕生日（2016年4月3日）、200ml献血が解禁されたその日から始まった献血生活が、本日ついに70回目を迎えまし...

♡ 83



Sho Sakai
2025/08/13 20:52

Bone Marrow Bank Youth Ambassador

2024 – present

I engage in activities to share information about hematopoietic stem cell transplantation and promote donor registration. If you're interested in inviting a speaker or organizing a promotional booth about the Japan Marrow Donor Program at your school, university, or organization, please reach out. We can coordinate with the Japan Marrow Donor Program directly. To increase the number of young bone marrow donors, I believe it is important to share the voices of those who have actually donated and their families.

Talks and media appearances

Sep. 4, 2026
Nursing school in Tokyo
Reaching younger generations about bone marrow donation: reflections from a talk

Apr. 14, 2026
FM Shonan Napassa, “Kochira Radio Bangumi Seisakubu”
Broadcast archive (Spotify)
Reaching younger generations about bone marrow donation: reflections from a talk

Feb. 14, 2026
Kamakura FM
I appeared on Kamakura FM as a bone marrow donor | Passing the baton of life begins with awareness

July 7, 2025

Nursing school in Tokyo

Gave a lecture as a bone marrow donor at a nursing school.

May 11, 2025

Health festival in Suginami

Participated in a health festival to promote the Japan Marrow Donor Program

Dec. 15, 2024

Egao Japan Marrow Donor Program Student Seminar (Ehime)

Gave a lecture as a bone marrow donor at the seminar

Dec. 2024

Japan Marrow Donor Program (interview, newsletter)

Interviewed on my donation experience

Articles and other notes

(Jan. 21, 2026) **No-blood 'swab registration' starts: data on the marrow bank and what we can do**

【採血不要】自宅からドナー登録できる「スワブ登録」が開始。データ…



はじめに どうも、Sho Sakaiです。私はこれまで、 「はじめての骨髄ドナー体験をすべて記録したリアルレポート」 や、献血に関する記事を通して…

♡ 34

Sho Sakai
2026/01/21 16:20

(Nov. 30, 2025) **An article about the background of “Bone Marrow Donya-” and my gacha experience**

可愛いだけじゃない。「骨髄ドニャー」誕生の背景とガチャ体験



マロニエゲート銀座で「骨髄ドニャー」というカプセルトイを見つけたので、思わず回してみました。 知り合いが働く病院にもこのガチャガチャはある...

♡ 21



Sho Sakai
2025/11/30 23:55

(May 28, 2025) **A summary of bone marrow transplant / bone marrow bank / blood donation related podcasts**

骨髄移植 / 骨髄バンク / 献血 関連 Podcast まとめ（2025/10/31更...



私が把握しているだけでもいくつかあるのでまとめました。それぞれ Spotify だけ添付していますが Apple Podcast などでも聞けると思います。 骨髄移...

♡ 12



Sho Sakai
2025/05/28 12:27

(Apr. 23, 2025) **"Voices of Donor Families" featured**

(Nov. 27, 2024) **The Japan Marrow Donor Program was registered as a T-ACT community group at the University of Tsukuba**

(Nov. 6, 2023) [A report on bone marrow donation experience and related information]

はじめての骨髄ドナー体験をすべて記録したリアルレポート（助成...

はじめに 1 つの記事で完結した内容としています。助成金や保険，公欠制度等様々な情報を「参考」以降で共有しているので，興味のある方はそちらだ...

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Sho Sakai
2023/11/06 10:34



QUALIFICATION / TEST SCORE

Mar. 2023 — Japanese High School Teacher's License (Information & Mathematics)

Mar. 2023 — Japanese Junior High School Teacher's License (Mathematics)

Mar. 2022 — TOEIC L&R score 735

Matsuo–Iwasawa Lab, Graduate School of Engineering, The University of Tokyo (Deep Learning courses)

Summer 2026 — Deep Generative Models 2026 Summer

Certificate

Spring 2026	Large Language Model 1 2026 Spring Certificate
Spring 2026	Fundamentals 2026 Spring Certificate
Spring 2026	Financial Market Trading and Machine Learning 2026 Spring Certificate

ACADEMIC SOCIETY

2025 – present	The Japanese Society for Artificial Intelligence
2023 – present	The Japan Statistical Society
Oct. 2023 – Mar. 2026	The Mathematical Society of Japan

PORTFOLIO

Reading notes: Claude and Finance articles ↗

Notes compiled while reading articles on Claude and finance.

Updated on Jul. 15, 2026

Master's Thesis / Hypothesis testing for PCR coefficients in high-dimensional data (JSS NEWS) ↗

Introduced in JAPAN STATISTICAL SOCIETY NEWS (April 20, 2025, No.203, Section 9: p.15).

Master's Thesis / Hypothesis testing for PCR coefficients in high-dimensional data (Mathematics Communication) ↗

Featured in Mathematical Society of Japan's 'Mathematics Communication' (Volume 30, No.1, May 2025) in the section '2024 Master's and Doctoral Theses'.

Graduation Thesis / Asymptotic Theory of Principal Component Analysis ↗

Related repository link available above.

Updated on Nov 23, 2024

Asymptotic Theory of PCA ↗

R

R code and mathematical notes on asymptotic properties of PCA under large-sample settings.

Updated on Nov 23, 2024

T.W. Anderson (2003) PCA Confirmatory Analysis ↗

R

Reimplementation of confirmatory PCA methods from Anderson's 2003 paper.

Updated on Nov 23, 2024

T.W. Anderson (2003) Hypothesis Testing ↗

R

Reproduction of hypothesis testing framework for PCA from Anderson (2003).

Updated on Nov 23, 2024

Principal Component Analysis ↗

Jupyter Notebook R

Notebooks to demonstrate PCA using both synthetic and real-world data. Implementations are available in Jupyter (Python) and R.

Updated on Nov 23, 2024

Conditional Probability and Multiplication Theorem ↗

LaTeX

A TeX document that explains the relationship between conditional probability and multiplication rule.

Updated on Nov 23, 2024

PRML with Python ↗

Jupyter Notebook

Selected implementations and exercises based on 'Pattern Recognition and Machine Learning' (Bishop).

Updated on Nov 1, 2023

Causal Inference ↗

Python Jupyter Notebook

Jupyter notebooks exploring potential outcome frameworks and causal graphs.

Updated on Jul 26, 2023

Programming with R, C, and Python ↗

R C Python

Introductory examples in multiple languages for basic computational logic and syntax.

Updated on Apr 19, 2023

Transformation of Matrix ↗

Utility scripts and explanations for matrix operations and transformations.

Updated on Feb 2, 2023

HP Last updated: November 4, 2026
CV Last updated: November 4, 2026

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